

BCE-C511
SEMESTER EXAMINATION-DECEMBER 2023
CLASS: BTech SEMESTER:V
COMPUTER NETWORK

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

SECTION-A
(SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: What is the function of data link control? How ENQ/ACK is different from POLL/SELECT method.

Question-2: What are different types of Error in data link layer and error control method. Explain with example.

Question-3: What is the different type of Cryptography Techniques?

Question-4: What is Network topology? Explain the different network topologies.

Question-5: What do you understand by Real Time Streaming Protocol (RTSP).

Question-6: What is routing .Explain the shortest path routing algorithm?

Question-7: Enumerate the mechanism of three way handshake protocol for TCP.

Question-8: A bit stream 1101011111is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted?

Question-9: Discuss the File transfer Protocol (FTP)with a neat diagram.

Question-10: Explain the different types of switching techniques and mention its advantages and disadvantages.

SECTION-B
(LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.
(4 X 10 = 40 Marks)

- Question-1: What are the different layers of OSI model? How they different from TCP/IP model.
- Question-2: What is meant by IP address? Why is it required? Explain the concept of class full and classless addressing.
- Question-3: Define routing and define distance vector routing and link state routing.
- Question-4: Explain the following WWW, HTTP, SMTP,SNMP.
- Question-5: A channel has a bit rate of 4 Kbps and one-way propagation delay of 30 msec. The channel uses stop and wait protocol. The transmission time of the acknowledgement frame is negligible. To get a channel efficiency of at least 75%, the minimum frame size should be.
- Question-6: Explain the difference between UDP &TCP.
- Question-7: Find the class of the following addresses, subnet mask, broadcastid, network id
1)227.13.14.88 2)10.32.65.269 3)191.25.85.36 4)241.96.35.46 5)242.12.13.1
- Question-8: Describe the role of a DNS on a computer network with reference to its components.